

2 Tilecal read-out system for the HL-LHC

The off-detector electronics of the upgraded TileCal will provide digitized signals at 40 MHz to the Level 0 trigger system through the Trigger and Data Acquisition interface, while the so-called Front-End Link eXchange (FELIX) system will read-out data stored in pipelines in Tile Preprocessors (TilePPr) at 1 MHz. The TilePPrs will interface the on- and off-detector electronics through multi-Gbps optical links. The power will be monitored and distributed to the on-detector electronics by the Detector Control System that interfaces with the Low Voltage Power Supplies in charge of distributing power to all the digital and analogue electronics, and the High Voltage Power Supplies that will provide high voltage to each PMT.

The on-detector is comprised by 896 Minidrawers (MDs), in which data from all TileCal PMTs will continuously be sampled at 40 MHz. Each MD will host up to twelve channels by means of twelve PMTs that turn light pulses to electric signals to be shaped and conditioned by Front-End Boards. A Mainboard (MB) will continuously sample and digitize two gains of shaped signals of the MD PMTs. A Daughterboard (DB) is in charge of distributing LHC synchronized timing, configuration and control to the front-end, and continuously transmitting the digital data from all the MB channels to the off-detector systems via multi-Gbps optical links.

3 ATLAS TileCal link Daughterboard

The DB is a radiation tolerant board with a redundancy layer of two functionally-equivalent halves, interconnecting the on- and off-detector by means of a 400-pin FMC connector and multi-Gbps optical links, respectively [3]. The DB, currently on its sixth revision (DB6, Figure 2), powers the optical links by means of four SFP+ modules that drive two downlinks and four uplinks. The board information received by the board from the off-detector by means of 4.8 Gbps GBT-FEC downlinks powered by CERN radiation hard GBTx ASICs includes clock signals, time synchronization and configuration commands.

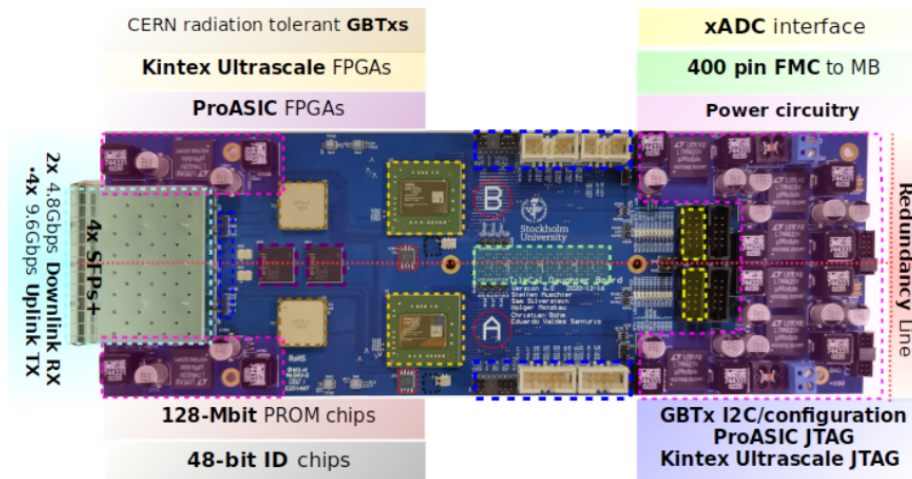


Figure 2. The DB6 layout.